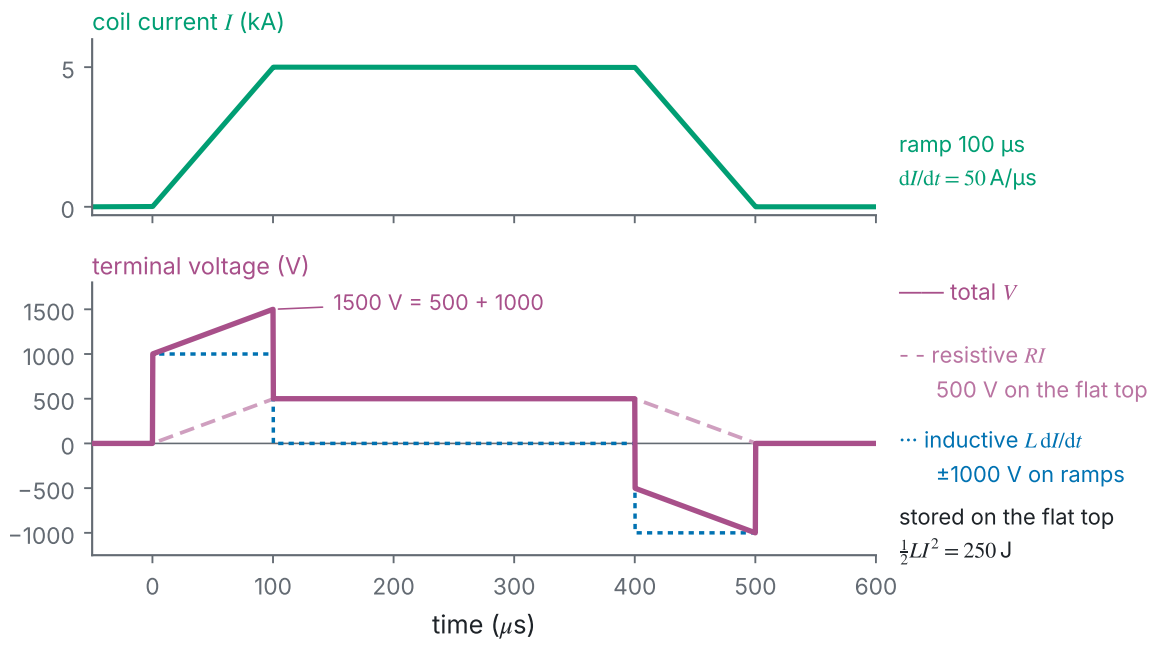
coil current I (kA)
5
0
ramp 100 µs
$\mathrm{d}I/\mathrm{d}t = 50$ A/µs
terminal voltage (V)
1500 V = 500 + 1000
1500
1000
500
0
−500
−1000
total V
- - resistive RI
500 V on the flat top
··· inductive $L\,\mathrm{d}I/\mathrm{d}t$
±1000 V on ramps
stored on the flat top
$\frac{1}{2}LI^2 = 250$ J
0
100
200
300
400
500
600
time (μs)